LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

ME-12-TYIN-007 | Yuqiao Li, Chun Yin Ku (John), Jiaquan Zhong | Supervisors: Jonathan Eden, Ying Tan

1. Introduction

1.1 Project background

Commercial models cannot meet specific research and development needs. 4DMedical's XV scanner enables research using our controller and the medical team's lung model. This stage validates one 250 mL branch prototype.

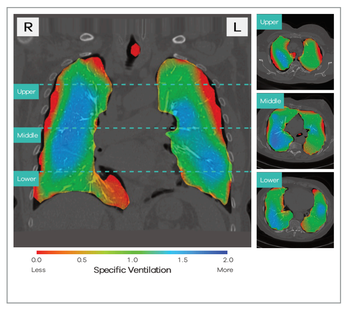


Fig. 1. XV LVAS ventilation map.

1.2 Project objective

- Adjust breathing rate and I:E.
- Control tidal volume.
- Program regional airflow.
- Support repeatable breathing cycles.
- UI for interfacing.

1.3 Nominal breathing reference

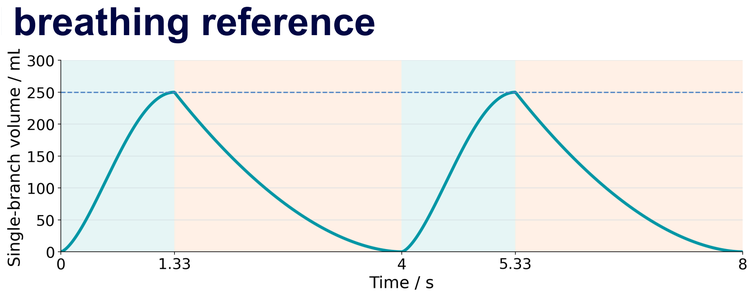


Fig. 2. Nominal breathing cycle: 15 bpm; I:E = 1:2.

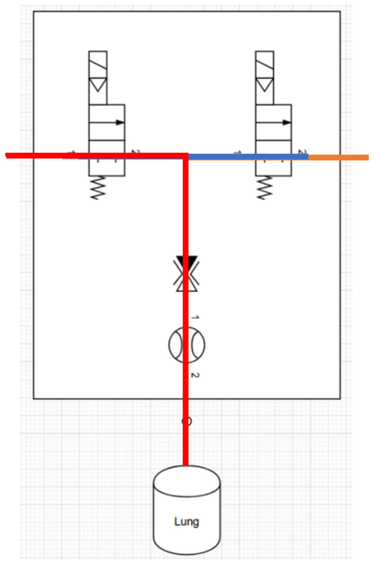
Respiratory rate	**12–16 bpm; nominal 15 bpm**
Inhale:Exhale time	**1:2; 1.33 s / 2.67 s at 15 bpm**
Tidal volume	**250 mL per branch**
Regional airflow	**Independently programmable by branch**

Nominal reference values shown; operating parameters are programmable.

Source: 4DMedical, XV LVAS Info Sheet, DOC-5070 US V1 (2023), pp. 1-2.

2. Pneumatic System & Hardware

2.1 Pneumatic system prototype



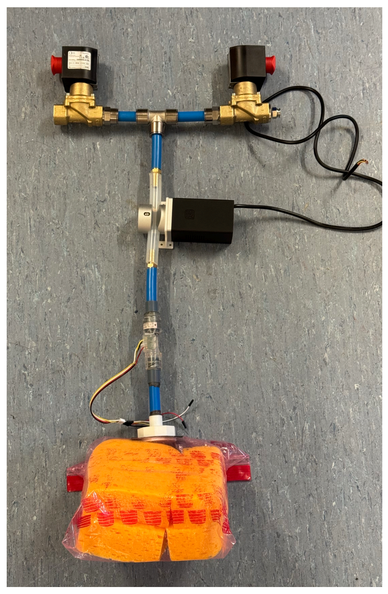


Fig. 3. Single-lung pneumatic route and 250 mL prototype.

2.2 Embedded control architecture

PC / Python UI — **MCU** — **24 V bridge PCB** — **Valves + sensor**

MCU executes commands from UI and acquires flow data. A custom bridging PCB integrates 24 V valve drivers, proportional-valve control circuitry, and sensor interfaces.

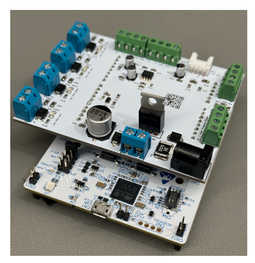


Fig. 4. Control architecture and bridge PCB connected to STM32.

3. Integrated Control System & Software

3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

3.2 Operator and developer interface

User Mode | **Developer Mode**

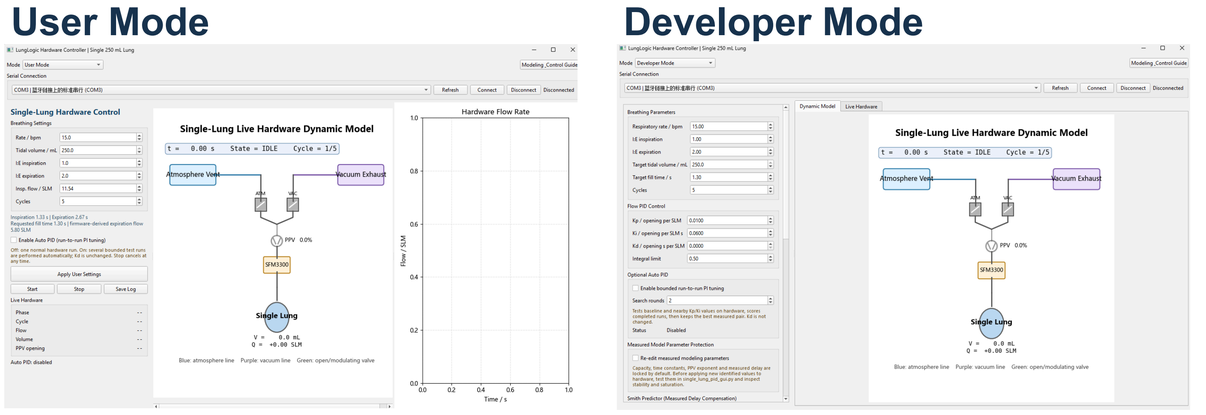


Fig. 5. User and Developer modes.

3.3 Model-assisted flow feedback

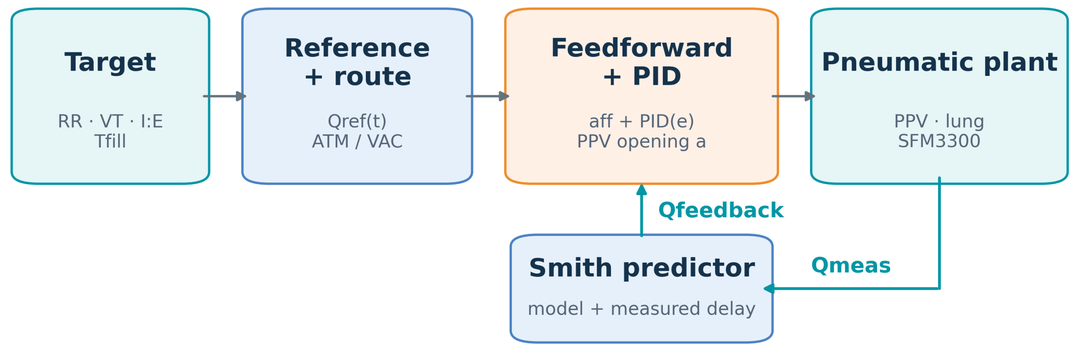


Fig. 6. Feedback control with optional Smith prediction.

3.4 Optional Auto PID and measured-delay compensation

Auto PID scores bounded Kp/Ki over full runs by flow, volume and saturation; optional Smith prediction uses the model and measured delay.

4. Results

4.1 Digital-twin simulation

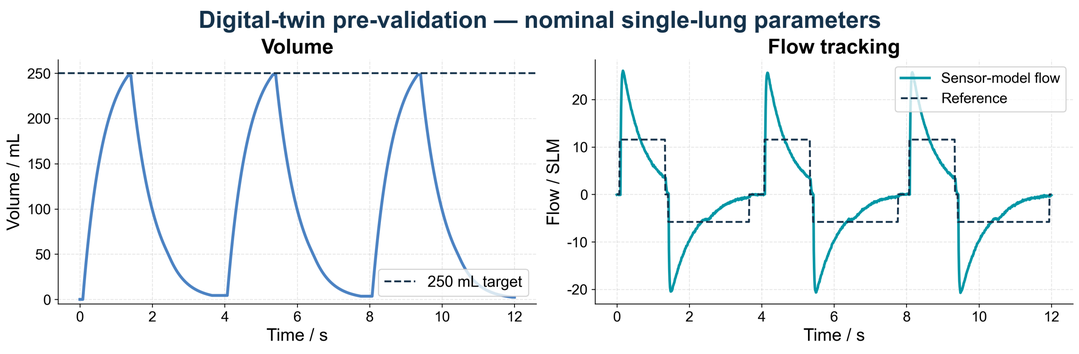


Fig. 7. Digital-twin volume and flow tracking at 15 bpm.

Open-loop tests identified distinct inspiration and expiration dynamics: plateau time constants of 2.53 s (ATM) and 1.31 s (VAC). Near closure, both responses slow. These are effective fits, not isolated valve delays.

4.2 Physical measurements

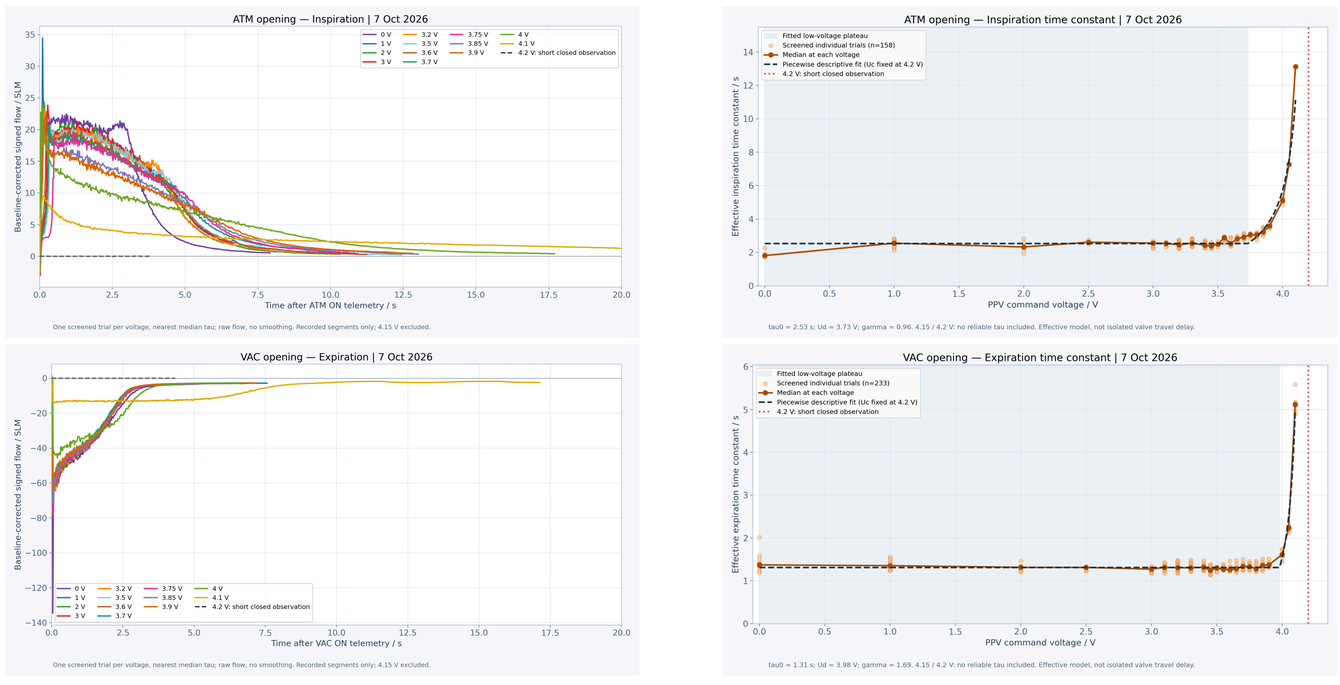


Fig. 8. ATM / VAC flow responses and effective time constants (7 Oct).

Industry collaborator: 4DMedical | Background: XV LVAS Info Sheet, DOC-5070 US V1 (2023) | Control: current single-lung system | Results: simulation + benchtop measurements | Bench research prototype.